4. Selenium Automation on Firefox

Step 1: Create a Maven Project (MyMavenSeleniumAppFireFox).

To create a basic Maven Project, use the following command:

mvn archetype:generate -DgroupId=com.example DartifactId=MyMavenSeleniumAppFireFox -DarchetypeArtifactId=maven archetype-quickstart -DinteractiveMode=false

```
shreya@ShreyaM:~$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp01 -DarchetypeArtifactId=maven-archetype-quickstart -Dint
eractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/shreya
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenSeleniumApp01
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/shreya/MyMavenSeleniumApp01
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  32.878 s
[INFO] Finished at: 2026-03-17T22:37:01+05:30
[INFO] ------------------------------------------------------------------------
```

Step 2 : edit the pom.xml file and add the dependencies

The pom.xml has the code

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
   xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">

   <modelVersion>4.0.0</modelVersion>
   <groupId>com.example</groupId>
   <artifactId>MyMavenExerciseAutomationApp</artifactId>
   <packaging>jar</packaging>
   <version>1.0-SNAPSHOT</version>
   <name>MyMavenExerciseAutomationApp</name>
   <url>http://maven.apache.org</url>

   <dependencies>
      <dependency>
         <groupId>junit</groupId>
         <artifactId>junit</artifactId>
         <version>3.8.1</version>
         <scope>test</scope>
      </dependency>

      <!-- https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->
      <dependency>
```

```
        <groupId>org.seleniumhq.selenium</groupId>
        <artifactId>selenium-java</artifactId>
        <version>4.29.0</version>
    </dependency>

    <dependency>
        <groupId>io.github.bonigarcia</groupId>
        <artifactId>webdrivermanager</artifactId>
        <version>5.8.0</version>
    </dependency>
  </dependencies>
<!-- Build: Configuring plugins and build settings -->
<build>
<plugins>
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.7</source>
<target>1.7</target>
</configuration>
</plugin>
<!-- Example: Maven Surefire Plugin to run tests -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>
<plugin>
        <groupId>org.apache.maven.plugins</groupId>

        <artifactId>maven-jar-plugin</artifactId>
        <version>3.0.1</version>
       <configuration>
          <archive>
             <manifestEntries>
                <Main-Class>com.example.App</Main-Class>
             </manifestEntries>
          </archive>
        </configuration>
    </plugin>
</plugins>
</build>
```

```
</project>
```

Step 3: Edit the App.java

```
package com.example;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.WindowType;

public class App {

    public static void main(String[] args) {

        WebDriver driver = null;

        try {
            FirefoxOptions options = new FirefoxOptions();
            options.addArguments("--disable-gpu");

            driver = new FirefoxDriver(options);


            // SauceDemo Login

            driver.get("https://www.saucedemo.com/");
            driver.manage().window().maximize();

            Thread.sleep(2000);

            driver.findElement(By.id("user-name")).sendKeys("standard_user");
            Thread.sleep(1000);

            driver.findElement(By.id("password")).sendKeys("secret_sauce");
            Thread.sleep(1000);

            driver.findElement(By.id("login-button")).click();
            Thread.sleep(3000);

            System.out.println(" SauceDemo login done");

            // AutomationExercise

            driver.switchTo().newWindow(WindowType.TAB);
            driver.get("https://automationexercise.com/");
            Thread.sleep(5000);

            try {
```

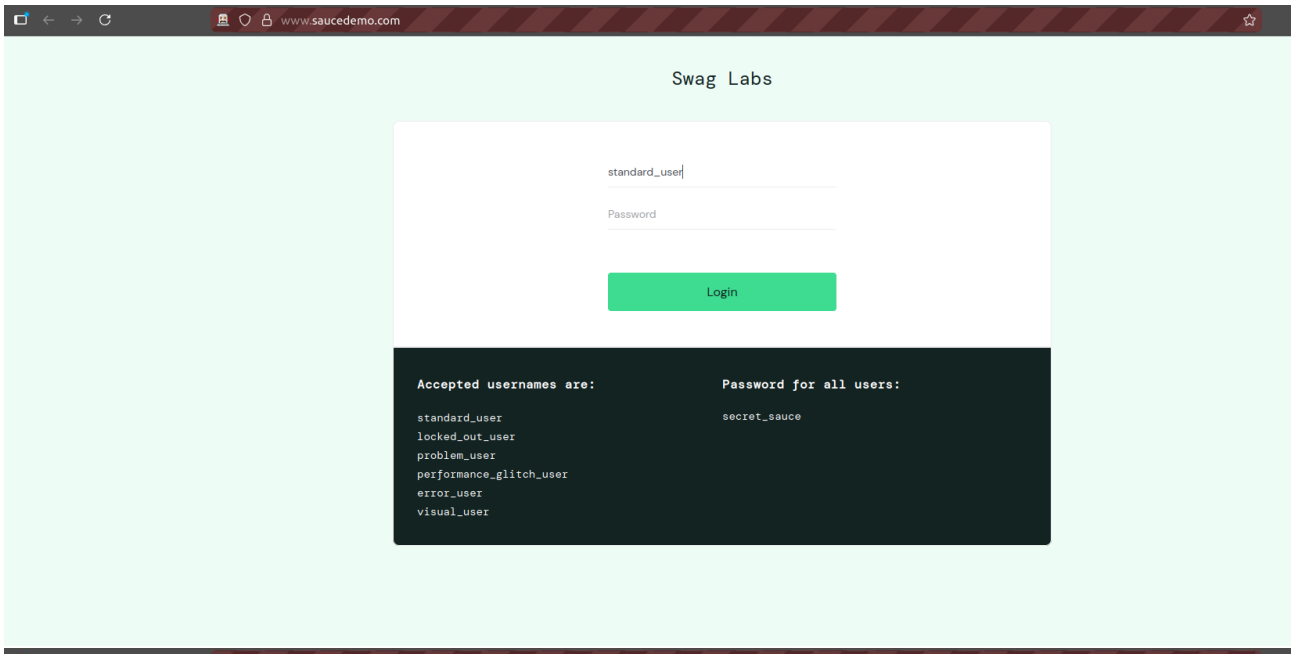
```
            driver.findElement(By.cssSelector(".grippy-host")).click();
        } catch (Exception e) {
            System.out.println("Ad popup not found");
        }

        Thread.sleep(2000);

        driver.findElement(By.xpath("(//a[contains(text(),'Add to cart')])[1]")).click();
        Thread.sleep(4000);

        driver.findElement(By.xpath("//button[contains(text(),'Continue Shopping')]")).click();

        System.out.println(" Product added to cart");

        // Practice Test Login

        driver.switchTo().newWindow(WindowType.TAB);
        driver.get("https://practicetestautomation.com/practice-test-login/");

        Thread.sleep(2000);

        driver.findElement(By.id("username")).sendKeys("student");
        Thread.sleep(1000);

        driver.findElement(By.id("password")).sendKeys("Password123");
        Thread.sleep(1000);

        driver.findElement(By.id("submit")).click();

        Thread.sleep(3000);

        System.out.println(" PracticeTest login done");

    } catch (Exception e) {
        System.out.println(" Error: " + e.getMessage());
    } finally {
        if (driver != null) {
            try {
                Thread.sleep(3000);
                driver.quit();
            } catch (Exception e) {
                System.out.println("Error closing browser");
            }
        }
    }
  }
}
```

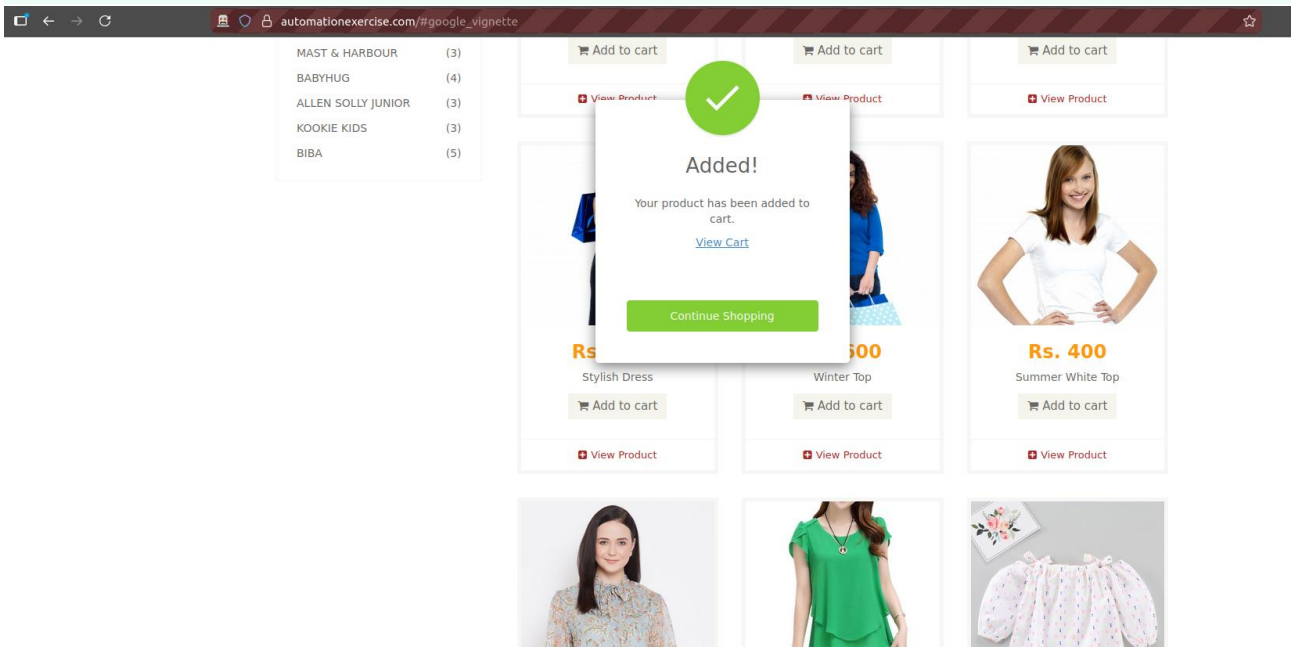
Step 4:build and run the maven application

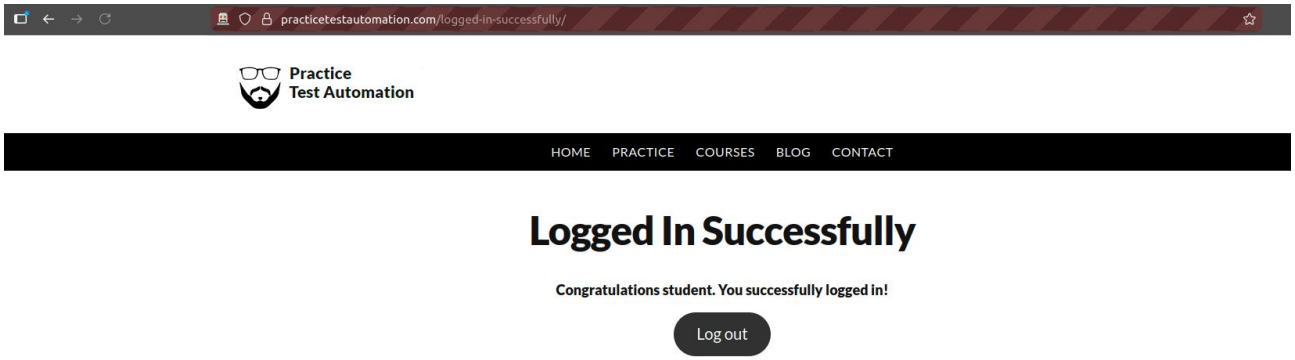
To execute type the command **mvn exec:java -Dexec.mainClass="com.example.App"**

```
shreya@ShreyaM:~/devops/firefox$ mvn exec:java -Dexec.mainClass="com.example.App"
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< com.example:MyMavenSeleniumApp01 >------------------
[INFO] Building MyMavenSeleniumApp01 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- exec-maven-plugin:3.6.3:java (default-cli) @ MyMavenSeleniumApp01 ---
✅ SauceDemo login done
✅ Product added to cart
✅ PracticeTest login done
```

Results:






practicetestautomation.com/logged-in-successfully/
Practice
Test Automation
HOME PRACTICE COURSES BLOG CONTACT
Logged In Successfully
Congratulations student. You successfully logged in!
Log out